

Table 7 - STAT 331/531 Final Project Report  
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**What skills are you intending to assess with your lab?**

This lab is intended to provide students with practice cleaning real field datasets, building simple functions to streamline making standard field measurements/conversions, and to visualize data in graphical and map formats. Students are provided with a real dataset, characterized by standard user errors associated with field data collection: missing values, typos, and inconsistencies. The student starts with two datasets that they need to combine. This is common in research, as methods change and update to accommodate new situations. Most libraries are unloaded from the start, forcing students to think about topics they've learned and consider what libraries are necessary to perform certain tasks.

The first portion of the lab is working to combine these two datasets and clean them in preparation for measurements, analysis, and visualization. The students are provided some guidelines, such as which columns to keep, but are prompted to use the appropriate function, such as the full join, data-time cleaning, and standard dplyr functions.

After the students have combined and cleaned the datasets (converted appropriate date-time values and cleaned site names), they are prompted to perform some data manipulation. Using the mutate() function, they are asked to make a new column of averages and remove rows. This prepares students to create a report standard table displaying the data of choice: adult frog measurements grouped by island and sex. Instead of manually doing a mutate for each standard body metric (sd, mean, max, min values), the students are asked to build a function to make the analysis more efficient. The function is designed to accomplish two tasks: 1.) group by user-specified variables in order to prep the data table, and 2.) calculate sd, mean, max, and min values from user input columns. This provides students with practice building functions that can be generalized to apply to future datasets.

Clean, organized, and effective data communication is a crucial skill in research and data communication. The students are asked to apply their function to help construct a nicely formatted table. We provide guidance for how the table should be designed (what message are we trying to communicate in the data), and students must design a table to match by specifying a table with appropriate group headers for specific rows and columns. The students are provided a table to match, but designing liberty is welcome in terms of font and package used (Using either gt() or kableExtra() is acceptable.)

Students also create a new function that converts Celsius to Fahrenheit. Even though the conversion is applied to only 2 columns, it encourages the practice of creating reusable functions in the future instead of repeating code. This can be used on new variables, as detailed in the lab; the function can eventually be applied to dorsal and cloacal temperature for future analysis.

Finally, students can visualize and display the very data they have been working with, in fancy graphs and detailed maps. By plotting the data, students are able to easily identify false/incorrect data. Creativity is encouraged for the user and helps them understand the importance of clear and legible graphics.

**What problems do you think would be easy for students to answer? What questions are difficult?**

Question 1 was intended to be quick and straightforward. However, it may be challenging for students who are unsure which data-cleaning functions are most appropriate for the task.

Questions 2, 3a, and 6 should be relatively easy because they rely on concepts and functions that students have practiced throughout the course.

Questions 4 and 5 were intentionally designed to be more difficult because they require students to create and apply custom functions. Since Question 5 builds directly on Question 4, difficulties in one question will likely affect performance on the other.

Question 7 is moderately difficult because it requires students to organize and communicate data effectively in a publication-quality table. The challenge is primarily in formatting and attention to detail rather than statistical complexity.

Question 8 may not be too hard, but it can be troublesome because it combines data visualization and mapping skills. Students must think carefully about how to present information clearly and effectively.

**Sources used**

- [Creating Date/Time objects with inconsistent formats](#)
- [Renaming Columns Headers](#)
- [Mapping California](#)